Precise Definitions and Counterexamples
Answer Key

High School Geometry   Definitions, Postulates, and Proof Logic

Quick Answers

1. -1, 5	4. 3	7. —
2. 5	5. $-\frac{3}{4}$	8. —
3. $-\frac{5}{4}$	6. -4	

1. Midpoint of $\overline{AB}$, $A(-6,2)$, $B(4,8)$.
Solution. The definition names the point with $AM = MB$, and the midpoint formula is that condition written in coordinates: $M = \left(\dfrac{-6+4}{2}, \dfrac{2+8}{2}\right) = \left(\dfrac{-2}{2}, \dfrac{10}{2}\right)$, so $\boxed{M = (-1,\ 5)}$
Why the student's wording fails: "near the middle" is not a test — $(-0.9, 5.1)$ is also near the middle, so the wording names a neighbourhood, not a point. A definition must single out exactly one object.

2. Is $\triangle PQR$ isosceles? $P(0,0)$, $Q(6,0)$, $R(3,4)$.
Solution. $PR = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{9+16} = \sqrt{25}$, so $\boxed{PR = 5}$ By symmetry $QR = \sqrt{(3-6)^2 + 4^2} = 5$ as well, so two sides are congruent and $\triangle PQR$ is isosceles (note $PQ = 6 \neq 5$, so it is not equilateral). *On the wording:* under "at least two congruent sides" every equilateral triangle is also isosceles; under "exactly two" it is not. Modern textbooks use "at least two" so that theorems about isosceles triangles apply to equilateral ones without a separate case — this is a choice a definition makes, and it must be made explicitly.

3. Kai's claim about the perpendicular bisector. $A(-4,1)$, $B(2,5)$, $M(-1,3)$, $C(3,-2)$.
Solution. Slope of $\overleftrightarrow{CM} = \dfrac{3-(-2)}{-1-3} = \dfrac{5}{-4}$, so $\boxed{-\frac{5}{4}}$ The product with the slope of $\overline{AB}$ is $\frac{2}{3} \cdot \left(-\frac{5}{4}\right) = -\frac{5}{6} \neq -1$, so $\overleftrightarrow{CM}$ is *not* perpendicular to $\overline{AB}$ even though it passes through the midpoint. *Conclusion:* Kai's statement is satisfied by infinitely many lines that are not the perpendicular bisector, so it fails job (2). The precise definition adds the missing condition: the line through the midpoint of $\overline{AB}$ *that is perpendicular to* $\overline{AB}$ (equivalently, the set of all points equidistant from A and B).

4. Converse of "square $\Rightarrow$ four right angles".
(a) Converse. "If a quadrilateral has four right angles, then it is a square."
(b) The computation. $QR = \sqrt{(6-6)^2 + (3-0)^2} = \sqrt{9}$, so $\boxed{QR = 3}$
(c) Why it settles the converse. $PQRS$ has four right angles, so it satisfies the hypothesis; but $PQ = 6$ and $QR = 3$ are not congruent, so it is not a square — the conclusion fails. One such object is enough: the converse is false. Note that the original statement stays true; a conditional and its converse are independent claims.

5. "Equal length ⇒ parallel".

Solution. Slope of $\overline{CD} = \dfrac{3-0}{-4-0} = \dfrac{3}{-4}$, so $\boxed{-\frac{3}{4}}$ $\overline{AB}$ has slope $\frac{4}{3}$ and $\overline{CD}$ has slope $-\frac{3}{4}$; unequal slopes mean the segments are not parallel (in fact the product is -1, so they are perpendicular — as far from parallel as possible). Both segments have length 5, so the hypothesis holds and the conclusion fails: a counterexample. Length says nothing about direction.

6. Find m so Line 2 is perpendicular to Line 1.

Solution. Slope of Line 2 is $\dfrac{m-0}{3-0} = \dfrac{m}{3}$. The definition of perpendicular gives the equation

$$\frac{3}{4} \cdot \frac{m}{3} = -1$$

$$\frac{m}{4} = -1 \qquad \Longrightarrow \qquad m = -4$$

$\boxed{m = -4}$ *Check:* Line 2 through $(0,0)$ and $(3,-4)$ has slope $-\frac{4}{3}$, and $\frac{3}{4} \cdot \left(-\frac{4}{3}\right) = -1$ ✓

7. Manual review — expected reasoning. The rhombus/kite question.

(a) $WXYZ$ has perpendicular diagonals ($\overline{WY}$ is vertical, $\overline{XZ}$ is horizontal), so it satisfies the student's statement. But its sides are 5, $\sqrt{45}$, $\sqrt{45}$, 5 — two different lengths — so it is a kite, not a rhombus. The statement therefore admits objects that are not rhombuses: it fails job (2).

(b) Precise definition: a rhombus is a parallelogram (equivalently, a quadrilateral) with four congruent sides. Accept any wording that is testable and excludes the kite, e.g. "a quadrilateral with all four sides congruent".

(c) The student's statement is a *property* (a theorem): every rhombus does have perpendicular diagonals. Properties run one way; definitions run both ways. Accept "theorem", "property", or "a true conditional whose converse is false", with the biconditional idea stated in some form. Do not accept an answer that merely repeats the definition without addressing what the student's statement is.

8. Manual review — expected reasoning. "Congruent diagonals ⇒ rectangle" is **false**.

Model counterexample. The isosceles trapezoid $A(0,0)$, $B(6,0)$, $C(5,3)$, $D(1,3)$. Diagonals: $AC = \sqrt{(5-0)^2 + (3-0)^2} = \sqrt{34}$ and $BD = \sqrt{(1-6)^2 + (3-0)^2} = \sqrt{34}$, so the diagonals are congruent and the hypothesis holds.

Which part of the definition fails. A rectangle is a quadrilateral with four right angles. Here $\overline{AD}$ runs from $(0,0)$ to $(1,3)$ with slope 3, while $\overline{AB}$ is horizontal (slope 0); $3 \cdot 0 \neq -1$, so $\angle A$ is not a right angle. The figure satisfies the hypothesis and fails the conclusion, so the claim is false.

Grading: accept any quadrilateral with congruent diagonals that is not a rectangle — every isosceles trapezoid works, and so does a suitable kite. The answer must (i) give specific vertices, (ii) show the two diagonal lengths are equal, and (iii) name the failing condition (a non-right angle, or non-parallel opposite sides). An answer that only says "a trapezoid" without coordinates or lengths is incomplete.